Golang Source Code File Naming Conventions

Summary

Golang is another name for, and refers to, the **Go Programming Language**.

This narrative describes the naming conventions which will be applied to ***Golang*** source code files containing types and methods, designed to perform a specific action, operation or calculation. These naming conventions describe a source code file hierarchy consisting of a top level source code file with public entry points and subsidiary levels containing source code files with private helper methods performing discrete elements, or components, of the specified function, operation or calculation.

Source Code File Naming Convention: Objectives and Structure

Overview

The Golang File Naming Convention is used to organize and group source code files containing types and methods related to the same action, operation or calculation. Source code files associated with a single action, operation or calculation are grouped and configured as a loose hierarchy using a file naming and numbering policy.

Source code files grouped together to perform a single action, operation or calculation are further subdivided into two classes: **Top Level Types and Methods** and **Supporting Types and Methods**.

Top Level Types and Methods represent the top level of the source code file hierarchy. **Top Level Types and Methods** contain entry points into the subject action, operation or calculation defined by the source code file hierarchy. **Top Level Types and Methods** usually contain public methods but, may also contain private methods as needed. **Top Level Types and Methods** source code file names consist of a **File Base Name** and a unique **File Numbering Component**.

Subsidiary or **Supporting Types and Methods** are configured in separate files using the same **File Base Name** component plus a unique **File Numbering Component** that designates their relative position in the overall source code file hierarchy. **Supporting Types** usually contain private methods but, may also contain public methods as needed.

Source Code File Name Components

All source code file names consist of three components:

1. **File Base Name**
2. **File Number Component**
3. **File Extension** (***.go***)

File Base Name

Prior to creating source code files and writing code, a common **File Base Name** is derived from the subject action, operation or calculation to be performed by this single group of source code files. The **File Base Name** will be included in all source files created in support of the subject action, operation or calculation.

Example:

Assume that a group of types and methods are designed to perform Natural log calculations using the arithmetic-geometric mean (a.k.a. AGM) technique. The **File Base Name** should therefore reflect the nature of the action, operation or calculation related to a single group of source code files.

File Base Name: `NaturalLogAGM`

File Number Component

The **File Number Component** consists of adding a trailing file number to the **File Base Name** using the format "**_0000**". File numbering components are unique within the source group hierarchy.

Both **Top Level Types and Methods** and **Supporting Types and Methods** use the same sequence of **File Numbering Components**.

Typically, source code files containing **Top Level Types and Methods** begin with numbering component "**_0010**" and increase in increments of ten (10) as new files are added.

Example:

```
[FILE BASE NAME]_0010.go # Top Level Types and Methods
[FILE BASE NAME]_0020.go # Supporting Types and Methods
```

Supporting Types and Methods are configured in source code files with numbering components typically beginning with "**_0020.go**", and increase in increments of ten (10) as new files are added.

Example:

```
[FILE BASE NAME]_0010.go # Top Level Types and Methods
[FILE BASE NAME]_0020.go # Supporting Types and Methods
[FILE BASE NAME]_0030.go # Supporting Types and Methods
[FILE BASE NAME]_0040.go # Supporting Types and Methods
[FILE BASE NAME]_0050.go # Supporting Types and Methods
```

This method of incrementing file numbers allows for the addition of new source code files as required by future modifications. If the numbering sequence becomes crowded, additional files may be inserted into the hierarchy using an extended numbering component formatted as "**_000.go**". Example:

```
[FILE BASE NAME]_0010.go
[FILE BASE NAME]_0020.go
[FILE BASE NAME]_0021.go
[FILE BASE NAME]_0021_010.go # Extended File Number Component
[FILE BASE NAME]_0022.go
[FILE BASE NAME]_0023.go
```

File Extension

All source code files are configured with a file extension of ***.go***. Example:

```
NaturalLogAGM_0010.go # Top Level Types and Methods
NaturalLogAGM_0020.go # Supporting Types and Methods
NaturalLogAGM_0030.go # Supporting Types and Methods
NaturalLogAGM_0040.go # Supporting Types and Methods
NaturalLogAGM_0050.go # Supporting Types and Methods
```

Top Level Types and Methods

Top Level Types and Methods are configured in one or more source code files named with the **File Base Name**, plus a unique **File Numbering Component** that designates their relative position in the overall source code file hierarchy.

Top Level Types and Methods represent the top level of the source code file hierarchy. **Top Level Types and Methods** contain entry points into the subject action, operation or calculation which defines the source code file group hierarchy.

Top Level Types and Methods usually contain public methods but, may also contain private methods as needed.

Top Level Type and Methods File Base Name Component

The File Base Name is derived from the purpose, objective or operational function of the source code file group, as explained above (File Base Name).

Top Level Type and Methods File Numbering Component

The **File Numbering Component** consists of adding a trailing file number to the **File Base Name** using the format "**_0000.go**". File numbering components are unique within the source group hierarchy. Typically, source code files containing **Top Level Types and Methods** begin with number component "**_0010.go**" and increase in increments of ten (10) as new files are added.

Example:

```
[FILE BASE NAME]_0010.go # Top Level Types and Methods
[FILE BASE NAME]_0020.go # Supporting Types and Methods
```

This method of incrementing file numbers allows for the addition of new source code files as required by future modifications. If the numbering sequence becomes crowded, additional files may be inserted into the hierarchy using an extended numbering component formatted as "**_000.go**".

Example: "***[FILE BASE NAME]_0030_010.go***"

Top Level Type and Methods: File Naming Example

- **Source Code Group Action, Operation or Calculation Objective:** Implement a group of types and methods designed to perform Natural log calculations using the arithmetic-geometric mean (a.k.a. AGM) technique.
- **File Base Name:** `NaturalLogAGM`
- **File Numbering Component:** `_0010.go`
- **Top Level Type and Methods Source Code File Name:** `NaturalLogAGM_0010.go`

Example:

```
NaturalLogAGM_0010.go # Top Level Types and Methods
NaturalLogAGM_0020.go # Supporting Types and Methods
NaturalLogAGM_0030.go # Supporting Types and Methods
NaturalLogAGM_0040.go # Supporting Types and Methods
NaturalLogAGM_0050.go # Supporting Types and Methods
```

Top Level Methods: Structure

Top Level Types usually contain public methods which serve as entry points into the subject action, operation or calculation. This policy is a general guideline and is not absolute. Top Level Types may contain private methods as dictated by operational circumstances or objectives.

Top Level Types and Methods: Call Tree

Top Level Types and Methods typically call lower level types and methods. This policy is a general guideline and is not absolute. **Top Level Types and Methods** may call other types and methods out of sequence as dictated by operational circumstances or objectives.

Supporting Types and Methods

Subsidiary or **Supporting Types and Methods** are configured in separate files with a **File Base Name** component and a unique **File Numbering Component** that designates the **Supporting Types and Methods** relative position in the overall source code file group hierarchy. Example: "**[FILE BASE NAME]_0020.go**"

File Base Name Component

The **File Base Name** is derived from the purpose, objective or operational function of the source code file group, as explained above (File Base Name).

File Number Component

The file numbering component consists of adding a trailing file number to the **File Base Name** using the format "**_0000.go**". File numbering components are unique within the source code file group hierarchy. Typically, source code files containing **Supporting Types and Methods** begin with number component "**_0020.go**" and increase in increments of ten (10) as new files are added.

Example:

```
[FILE BASE NAME]_0010.go # Top Level Types and Methods
[FILE BASE NAME]_0020.go # Supporting Types and Methods
[FILE BASE NAME]_0030.go # Supporting Types and Methods
[FILE BASE NAME]_0040.go # Supporting Types and Methods
[FILE BASE NAME]_0050.go # Supporting Types and Methods
```

This method of incrementing file numbers allows for the addition of new source code files as required by future modifications. If the numbering sequence becomes crowded, additional files may be inserted into the hierarchy using an additional numbering component formatted as "**_000.go**".

Example:

```
[FILE BASE NAME]_0010.go
[FILE BASE NAME]_0020.go
[FILE BASE NAME]_0021.go
[FILE BASE NAME]_0021_010.go # Extended File Number Component
[FILE BASE NAME]_0022.go
[FILE BASE NAME]_0023.go
```

Supporting Types and Methods Source Code: File Naming Example

- **Source Code Group Action, Operation or Calculation Objective:** Implement a group of types and methods designed to perform Natural log calculations using the arithmetic-geometric mean (a.k.a. AGM) technique.
- **File Base Name:** `NaturalLogAGM`
- **File Numbering Component:** `_0020.go`
- **Supporting Types and Methods Source Code File Name:** `NaturalLogAGM_0020.go`

Example:

```
NaturalLogAGM_0010.go # Top Level Types and Methods
NaturalLogAGM_0020.go # Supporting Types and Methods
NaturalLogAGM_0030.go # Supporting Types and Methods
NaturalLogAGM_0040.go # Supporting Types and Methods
NaturalLogAGM_0050.go # Supporting Types and Methods
```

Supporting Types and Methods: Structure

Supporting Types usually contain private methods called by higher level types and methods. This is the preferred format for **Supporting Types**. However, **Supporting Types**, may also contain public methods as required by operational circumstances or objectives.

Supporting Types and Methods: Call Tree

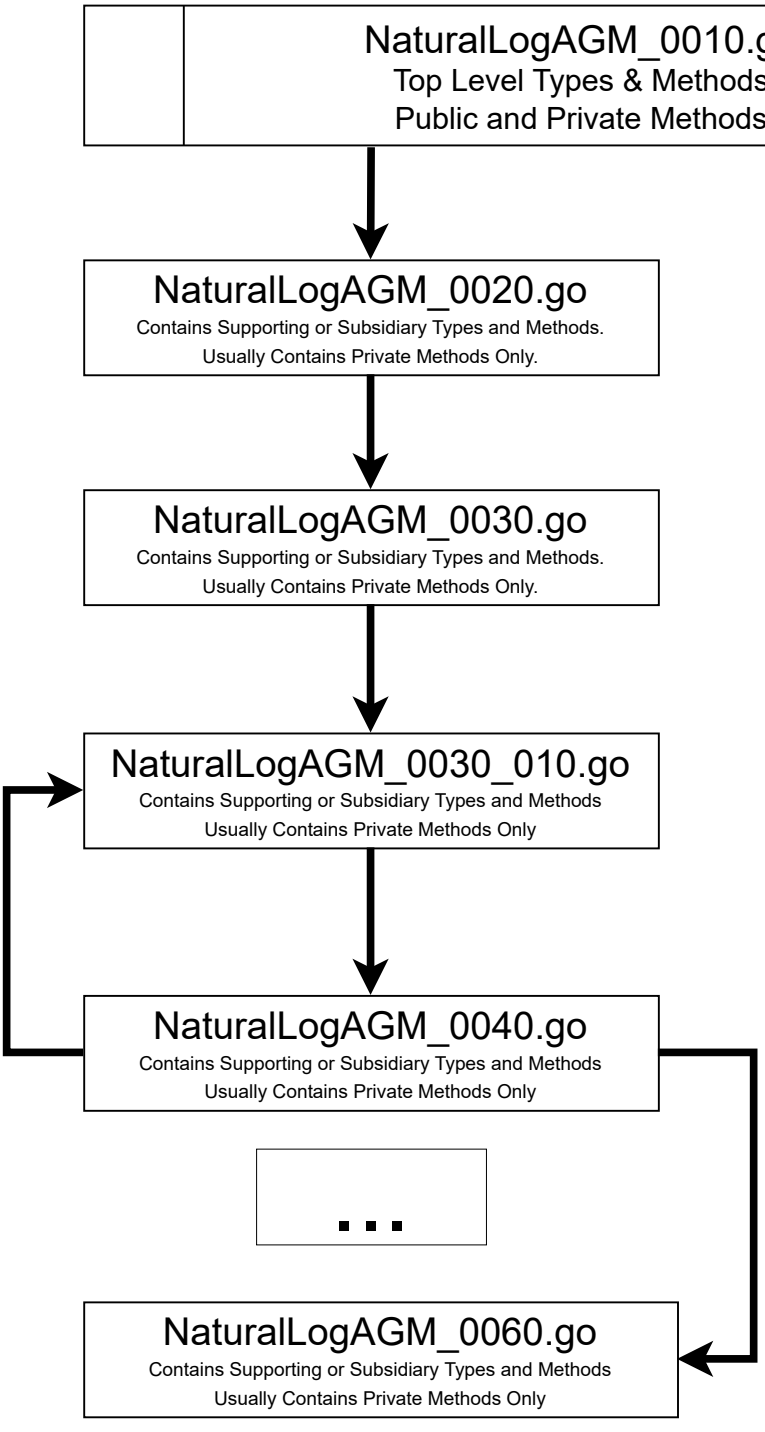
Supporting Types and Methods typically call lower level types and methods. This policy is a general guideline and is not absolute. **Supporting Types and Methods** may call other types and methods out of sequence as required by operational circumstances or objectives.

Golang Source Code File Naming Convention

This file naming convention is used to organize and group source code files containing custom types and methods related to the same action, operation or calculation.

Example File Base Name: NaturalLogAGM

Files with this base name perform Natural Log calculations using the arithmetic–geometric mean (a.k.a. AGM) technique. The File Base Name defines the action, operation or caluclation related to this group of files.



Typically, Higher Level Types and Methods call Lower Level Types and Methods.

Typically, Type Names are derived from file names.

```
type NaturalLogAGM_0020 struct {}
```

Over time, it may be necessary to insert source code files between initial sequence members. In these cases, add "_000.go" extended sequence designators, if standard numbering is not available.

Types and Methods may call other Types and Methods out of sequence. However, the preference is Higher Level Types and Methods call Lower Level Types and Methods.